

ZEROth REVIEW - PROJECT PHASE I

Explainable AI-Based Personal Financial Copilot

Predictive Budgeting, Goal Forecasting, and Financial Risk Analysis

Doron	Linton	Prince
Baskar	Aadhithya	
Kannapiran - 23cu0330020		

Introduction

- Personal finance management is a daily challenge — tracking expenses, planning savings, and achieving long-term financial goals
- Existing budgeting apps mostly record past transactions and generate basic reports
- They rarely help users understand their future financial situation
- Advances in AI, ML, Explainable AI (XAI), and LLMs open the door to smarter, more intelligent financial assistants
- This project proposes an Explainable AI-Based Personal Financial Copilot — combining predictive budgeting, goal forecasting, and risk analysis with transparent, AI-driven guidance



Motivation

- Financial planning is a real-world problem faced by almost everyone — students, employees, freelancers, and families alike
- Current apps offer predictions and suggestions without explaining why — leading to low user trust
- There is a clear gap between "tracking money" and "understanding & planning money"
- Explainable AI can bridge this gap — making financial recommendations transparent and understandable
- Working on this project strengthens practical knowledge in ML, XAI, Predictive Analytics, and FinTech, while solving a problem with genuine social impact



The Problem with Current Applications



Basic Tracking

Most applications only record past transactions and generate simple retrospective reports, completely lacking proactive financial planning.



Black-Box AI

Users receive automated predictions without understanding why they were generated, significantly reducing trust in AI-based financial systems.



Weak Forecasting

Existing systems offer limited financial risk analysis and struggle to provide accurate, personalized long-term financial goal forecasting.

Proposed Solution

- An AI-powered Copilot that moves beyond traditional budgeting to offer **proactive decision support**.
- **Predictive Budgeting:** Analyzes historical spending patterns using ML to accurately forecast future budgets.
- **Explainable AI (XAI):** Transparently explains the reasoning behind every prediction and recommendation to build user trust.
- **Conversational UX:** Allows users to interact naturally with their data for personalized, on-demand financial guidance.



Project Objectives

1. Core AI Development

Develop an intelligent personal financial management system that leverages machine learning techniques to predict future budgets and forecast goal achievement based on specific user spending behaviors.

2. Trust & Transparency

Analyze financial risks such as overspending and integrate Explainable AI (XAI) frameworks. The primary goal is to improve transparency, user trust, and financial literacy through a conversational interface.

Scope & Target Audience



Empowering Diverse Demographics

The proposed system is intended for **all age categories people** who wish to fundamentally improve their financial planning and decision-making.

The core scope includes comprehensive expense management, predictive budgeting, goal forecasting, and interactive conversational support driven by Explainable AI.

Expected Contributions



XAI Framework

Introducing an Explainable AI framework specifically designed for transparent, understandable financial recommendations.



Decision Support

Delivering improved financial decision support through highly personalized recommendations and holistic risk analysis.



Conversational UX

Deploying a conversational AI assistant that allows for natural, interactive, and accessible financial guidance.

Future Scope

-  **Open Banking APIs** Seamless integration for real-time, automated transaction fetching.
-  **Voice-Based Assistants** Enabling hands-free financial queries, inputs, and accessibility.
-  **Blockchain Technology** Implementing decentralized ledgers for highly secure financial records.
-  **Federated Learning** Privacy-preserving AI training models built directly on decentralized user data.

Review of Literature (Base Papers)

Paper Title / Source	Core Focus Area	Year
Smart Expense Tracker with Spending Pattern Prediction	Predictive Analytics & Budgeting	2026
arXiv:2406.11903	AI-based Personal Finance	2024
IEEE Paper - 11506666	Explainable AI (XAI) in Finance	TBD
Springer - 10.1007/s42521-025-00146-3	Intelligent Financial Assistance	2025
SSRN - 6615358	Financial Technology & ML	TBD

Project Execution & Planning



Thank You